

QRM II Graded Assignment (12)

Period 1, 2026

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0 Introduction

This assignment is to be completed in groups of 4-5. Further, all students in your group need to be assigned to the same R tutorial group (Friday's tutorial). You can sign yourself up for a group on Canvas. Please do so **before the start of your first R tutorial on Friday September 4th**. You can use the Discussion Board in Canvas if you do not have a group yet or if your group is incomplete.

The assignment has 5 parts, and each part corresponds to the course material of that week (with the exclusion of week 6, for which there is no R programming material).

You are supposed to hand in these assignments on Canvas at the following dates:

- **Deadline 1** *Sunday September 27th, at 23:59pm*: you are supposed to hand in weeks 1, 2, and 3 of this assignment. This will determine 18% of your overall course grade
- **Deadline 2** *Sunday October 11th, at 23:59pm*: you are supposed to hand in weeks 4, and 5 of this assignment. This will determine 12% of your overall course grade

The R tutorials (each Friday) will consist of two halves. During the first half, you will discuss the tutorial exercises. These can be downloaded separately from Canvas. During the second half, you can work on this graded assignment within your own group. The purpose is that you find out how to work with R for doing statistical analyses by yourself. The tutorial exercises are meant to teach you basic commands to get you started, but to answer the problem sets in this assignment, you might need to research your own solutions, and use functions and commands not described in the tutorial exercises. Learning how to solve your own research problems is integral part of learning R. When you and your group get stuck on how to approach an exercise, the hierarchy in finding your way is as follows:

- use the concepts from the tutorial exercises;
- use the cheat sheets available on Canvas;
- use Google, YouTube, StackOverflow, or another website;
- ask the teacher.

The use of generative AI is **not** permitted and may result in a grade of 0. See the AI protocol in the course manual for details.

To answer the assignment, you can simply fill out this R markdown document. There are designated places which you can fill with R code. There are also designated spaces for you to answer each question. Often, the structure of an answer will be as follows. First, you type the R code in the designated box. This will show how you analyzed the data to get the answer to the question. Below the box for the R code, you will then summarize your answer to the question, i.e. what are the conclusions that you draw from the data analysis?

When handing in, you are supposed to submit this .Rmd file, and a knitted version of this document. You can knit this document to pdf, word, or html. Knitting to pdf requires you to have a .tex distribution installed on your computer. Knitting to Word requires you to have Word installed.

The exercises are designed such that you should be able to finish the majority of them during the tutorial each week. If you are not able to finish them fully during that time, you are expected to work on it in your own time using the computers on campus or your own device. It is best to meet as a group in-person when working together. If you want to work remotely, github is a good platform to guarantee smooth collaboration. Alternatively, you can email this .Rmd file back and forth to one another as a group, but this is not recommended as it is more cumbersome.

We encourage you to keep your code blocks, printing statements, and final answers, as short as possible. In any case, there is a page limit of 6 pages per week, which encompasses the total length of this document which consists of the questions, your coding lines, and your answers. When your answers to questions of the respective week exceed this page limit, they will not be graded, resulting in zero points.

```
movies2 <- read.delim("movies2.tsv", header = TRUE, sep = "\t",
                      quote = "\"", stringsAsFactors = FALSE, na.strings = "NA")
```

1 Week 1

1. Find the dataset “movies2.tsv” on Canvas. Describe your data set: How many observations does it have. How many variables are there? How many subjects? What consists of a subject? **[4 points]**

```
# Number of observations (rows)
nrow(movies2)
```

```
## [1] 606
```

```
# Number of variables (columns)
ncol(movies2)
```

```
## [1] 19
```

```
# Dimensions: observations x variables
dim(movies2)
```

```
## [1] 606 19
```

```
# Look at the variable names
names(movies2)
```

```
## [1] "index"          "budget"         "keywords"
## [4] "original_language" "title"          "popularity"
## [7] "release_date"    "revenue"        "runtime"
## [10] "vote_average"    "vote_count"     "genre"
## [13] "release_year"    "release_month"  "release_day"
## [16] "first_actor"     "first_actor_gender" "director_first_name"
## [19] "director_gender"
```

How many observations? 606 observations. How many variables? 19 variables. How many subjects? 606 subjects. What consists of a subject? One subject is one movie. Each row in the dataset represents a single movie.

2. Which of the following types of variables are present in your data set? (i) nominal; (ii) ordinal; (iii); continuous; (iv) discrete. If present, name one example of such a variable present in your data set. [4 points]

```
str(movies2)
```

```
## 'data.frame':    606 obs. of  19 variables:
## $ index          : int  3793 3853 2476 491 540 3238 306 1594 3396 3267 ...
## $ budget         : num  4.0e+06 3.5e+06 0.0 1.3e+08 7.5e+07 8.0e+06 1.0e+08 4.0e+07 7.0e+06 8.0e+06 ...
## $ keywords       : chr   NA "thriller forensic" "poison prison loss of family loss of father foster ...
## $ original_language : chr   "en" "en" "en" "en" ...
## $ title          : chr   "The Velocity of Gary" "2:13" "White Oleander" "Pompeii" ...
## $ popularity      : num   0.409 1.272 9.424 50.562 10.605 ...
## $ release_date    : chr   "1999-07-16" "2009-04-25" "2002-10-11" "2014-02-18" ...
## $ revenue         : num   0.00 0.00 0.00 1.18e+08 5.11e+07 ...
## $ runtime         : int   100 96 109 105 116 102 102 77 170 97 ...
## $ vote_average    : num   4.2 4.6 7.2 5.2 5 7.5 5.3 7.2 6.7 4.6 ...
## $ vote_count      : int    5 10 108 1267 166 1686 381 1894 41 30 ...
## $ genre           : chr   "Drama" "Thriller" "Drama" "Drama" ...
## $ release_year    : int   1999 2009 2002 2014 2003 2006 2009 2005 2013 1996 ...
## $ release_month   : int    7 4 10 2 6 7 6 9 2 5 ...
## $ release_day     : int   16 25 11 18 9 26 5 9 27 3 ...
## $ first_actor     : chr   "Vincent D'Onofrio" "Mark Thompson" "Alison Lohman" "Kit Harington" ...
## $ first_actor_gender : chr   "male" "male" "female" "male" ...
## $ director_first_name : chr   "Dan" "Charles" "Peter" "Paul" ...
## $ director_gender  : chr   "male" "male" "male" "male" ...
```

```
sapply(movies2, function(x) length(unique(x)))
```

```
##           index           budget           keywords           original_language
##           606             135             533             17
##           title           popularity           release_date           revenue
##           606             606             572             420
##           runtime           vote_average           vote_count           genre
##           98              49             425             15
##           release_year           release_month           release_day           first_actor
##           27              12             31             460
## first_actor_gender director_first_name           director_gender
##           3              302             3
```

Three of the four types of variables are present in the data set.

Nominal: Yes. For example, genre is nominal because its categories, such as Drama, Comedy, and Action, have no natural ordering.

Ordinal: No. There are no variables consisting of ordered categories in the data set.

Continuous: Yes. For example, revenue is a continuous variable because it can take many possible numerical values over a range.

Discrete: Yes. For example, `vote_count` is discrete because it represents a count and can only take whole-number values.

3. A movie studio wants to know which types of movies give maximal revenue. Perform the following steps to provide the movie studio with an analysis which corresponds to their request:
 - a. Summarize the variable revenue and report its mean, median, maximum, and minimum. **[2 points]**
 - b. Which movie has the highest revenue in your data set and how much is this revenue. Which movie has the lowest and how much is its revenue? If multiple movies have the exact same highest or lowest revenue, give only one example. **[2 points]**
 - c. Create a histogram of the variable revenue. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Describe the shape of the histogram. What does this tell you about the nature of making money in the movies industry? **[2 points]**
 - d. Add a new variable to your data set: the log of revenue. When creating this variable, what happens to movies for which revenue is zero? What then happens when you calculate the mean of log revenue? **[2 points]**
 - e. For movies that have a revenue of zero, replace log of revenue with "NA". What is now the mean of log of revenue? Create a histogram for log revenue, again with an appropriate title, x- and y-axis labels. Describe the shape of this histogram. **[2 points]**
 - f. For the variables revenue and log of revenue calculate their skewness and kurtosis. How do these numbers relate to the shape of the histograms that you estimated under d.) and e.)? **[2 points]**

For each step, you should provide first all the code you used to answer the question and then formulate an answer using full sentences.

Each week consists of 1, 2, or 3 subquestions. The total amount of points you can earn per week is 20 points.

Step a

```
# Summary statistics for the revenue variable
mean_rev <- mean(movies2$revenue, na.rm = TRUE)
median_rev <- median(movies2$revenue, na.rm = TRUE)
max_rev <- max(movies2$revenue, na.rm = TRUE)
min_rev <- min(movies2$revenue, na.rm = TRUE)
cat("Mean:", mean_rev, "\nMedian:", median_rev, "\nMax:", max_rev, "\nMin:", min_rev)
```

```
## Mean: 84953602
## Median: 16624122
## Max: 1506249360
## Min: 0
```

The revenue variable ranges from a minimum of \$0 to a maximum of \$1,506,249,360. The mean revenue is \$84,953,602, while the median is only \$16,624,121.5. The fact that the mean is more than five times the median shows that revenue is very unevenly distributed: Most movies earn well below the average, while a small number of blockbuster films make extremely high revenues and pull the mean upward.

Step b

```
movies2[which.max(movies2$revenue), c("title", "revenue")]
```

```
##           title      revenue
## 272 Furious 7 1506249360
```

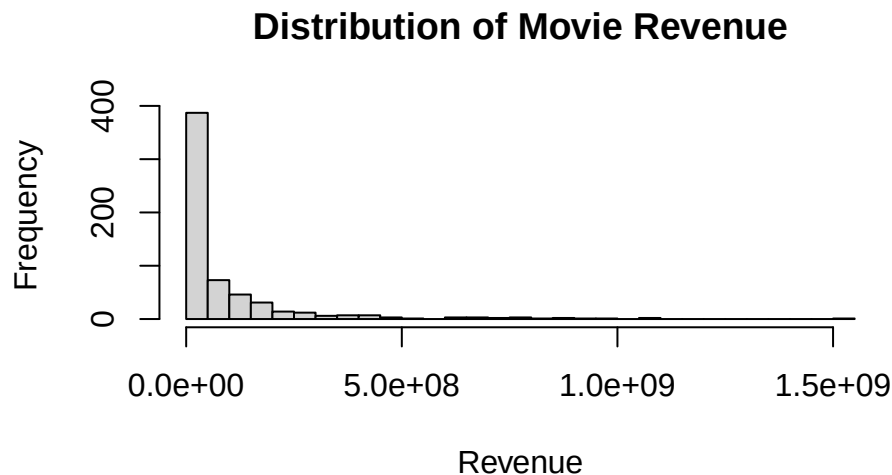
```
movies2[which.min(movies2$revenue), c("title", "revenue")]
```

```
##           title revenue
## 1 The Velocity of Gary      0
```

The movie with the highest revenue is Furious 7, with \$1,506,249,360. The lowest recorded revenue is \$0, and The Velocity of Gary is one example. However, 187 of the 606 movies have revenue recorded as \$0, so `which.min()` just returns the first one it finds.

Step c

```
hist(
  movies2$revenue,
  main = "Distribution of Movie Revenue",
  xlab = "Revenue",
  ylab = "Frequency",
  breaks = 30
)
```



The histogram of revenue is strongly right-skewed. Most movies are concentrated near \$0 revenue, while only a small number of movies have very high revenues. Around 387 of the 606 movies are in the first bin, while only about 20 movies earn more than \$500 million. A few blockbusters, such as Furious 7, are far out in the right tail.

This shows that movie revenues are very uneven. Most movies earn relatively little, while a small number of successful movies earn extremely high amounts. Because of these blockbusters, the mean revenue is much higher than the median, so the median gives a better idea of what a typical movie earns.

Step d

```
movies2$log_revenue <- log(movies2$revenue)
head(movies2[movies2$revenue == 0, c("revenue", "log_revenue")])
```

```
##    revenue log_revenue
## 1      0      -Inf
```

```
## 2      0      -Inf
## 3      0      -Inf
## 10     0      -Inf
## 16     0      -Inf
## 18     0      -Inf
```

```
mean(movies2$log_revenue)
```

```
## [1] -Inf
```

When `log()` is applied to revenue, the 187 movies with revenue equal to \$0 become `-Inf` in R, not `NA`. This is because the logarithm of 0 is not defined and approaches negative infinity.

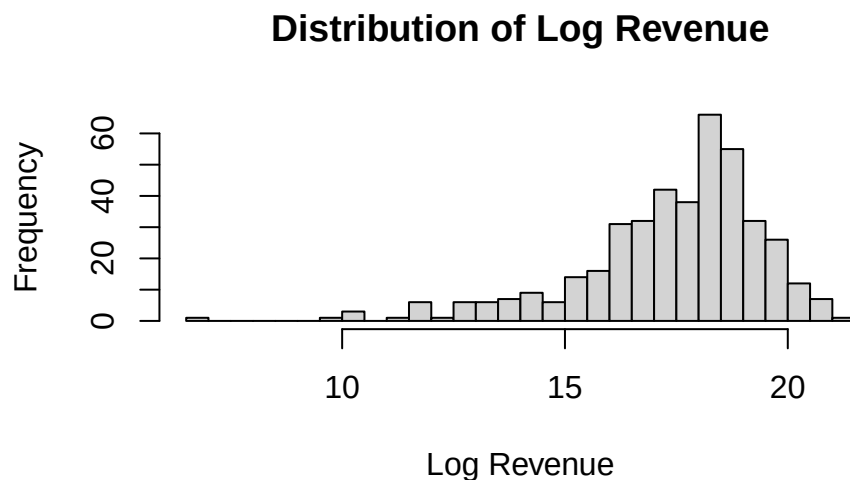
As a result, the mean of `log_revenue` is also `-Inf`. Even one `-Inf` value makes the overall mean `-Inf`, so this mean cannot be meaningfully interpreted.

Step e

```
movies2$log_revenue <- ifelse(
  movies2$revenue == 0,
  NA,
  log(movies2$revenue)
)
#calculate mean
mean(movies2$log_revenue, na.rm = TRUE)
```

```
## [1] 17.40298
```

```
#create hist
hist(
  movies2$log_revenue,
  main = "Distribution of Log Revenue",
  xlab = "Log Revenue",
  ylab = "Frequency",
  breaks = 30
)
```



After replacing the `-Inf`

values for movies with zero revenue with NA, the mean of `log_revenue` is approximately 17.40. R now calculates the mean using the 419 movies with positive revenue.

The histogram of `log_revenue` is much more symmetric than the original revenue histogram. Instead of having a long right tail, most values are now concentrated around 17–19, with a slightly longer tail on the left. The distribution is therefore much closer to a bell shape, although it is still slightly left-skewed.

The log transformation reduces the effect of extremely high revenues and spreads out the lower values. This makes the distribution more balanced and reduces the influence of blockbuster movies.

Step f

```
skewness <- function(x) {  
  x <- na.omit(x)  
  n <- length(x)  
  m3 <- sum((x - mean(x))^3) / n  
  s3 <- (sum((x - mean(x))^2) / n)^(3/2)  
  return(m3 / s3)  
}  
kurtosis <- function(x) {  
  x <- na.omit(x)  
  n <- length(x)  
  m4 <- sum((x - mean(x))^4) / n  
  s4 <- (sum((x - mean(x))^2) / n)^2  
  return(m4 / s4) # Pearson kurtosis (normal distribution = 3)  
}  
skew_rev <- skewness(movies2$revenue)  
kurt_rev <- kurtosis(movies2$revenue)  
  
skew_log <- skewness(movies2$log_revenue)  
kurt_log <- kurtosis(movies2$log_revenue)  
  
cat("Raw Revenue -> Skewness:", skew_rev, "| Kurtosis:", kurt_rev, "\n")
```

```
## Raw Revenue -> Skewness: 3.745914 | Kurtosis: 20.75961
```

```
cat("Log Revenue -> Skewness:", skew_log, "| Kurtosis:", kurt_log, "\n")
```

```
## Log Revenue -> Skewness: -1.316195 | Kurtosis: 5.578622
```

The skewness of raw revenue is approximately 3.75. Since this is a large positive value, it confirms that revenue is strongly right-skewed, with a long tail toward very high revenues. This also explains why the mean is higher than the median. The kurtosis is about 20.76, which is much higher than 3, the kurtosis of a normal distribution. This shows that the distribution has a sharp peak and several extreme values, such as blockbuster movies.

For `log_revenue`, the skewness is approximately -1.32 . This means the distribution is slightly left-skewed, which matches the histogram from part e. The mean of `log_revenue` (17.40) is also slightly below the median (17.88). Its kurtosis is around 5.58. This is still higher than 3, so the distribution is still more peaked than a normal distribution.

Overall, the log transformation greatly reduces both the skewness and kurtosis of revenue. The distribution becomes more symmetric and less affected by extreme values, although it is still not perfectly normal.

2 Week 2

1 Is your dataset `movies2.tsv` the full population, or is it a sample of a larger population? If the latter, how would you describe the full population? [4 points]

The `movies2.tsv` dataset is a sample of a larger population, the whole population consists of all movies that have the same selection criteria as the movies included in this dataset.

2 For this question, you will assume that your data set is the full population.

- What is the mean of the variable `vote_average` in your data set? [2 points]
- Create a new data set, called `movies_sample`. Make sure that it is a random sample of your data set of 25 movies. What is the mean of `vote_average` in this random sample? How does it compare to the mean of `vote_average` in 2a? [2 points]
- In a for loop, create 100 different samples of 25 movies, as in b, and estimate the mean within each sample. Save the mean of each sample in a vector called `sample_means`. So the first position of the vector would have the mean of the first sample, the second position the mean of the second sample, etc. Print the start of this vector. [2 points]
- Summarize and make a histogram of `sample_means`. What is the mean, standard deviation and shape of its distribution? [2 points]
- Using the formulas of the *Central Limit Theorem*, give the distribution of the sample mean of `vote_average` in a sample of 25 movies from your population (including its expected value and standard deviation). How do these values compare to your answers at d.)? [2 points]

Step a

```
population_mean <- mean(movies2$vote_average, na.rm = TRUE)
population_sd   <- sd(movies2$vote_average, na.rm = TRUE)
population_mean
```

```
## [1] 6.047855
```

```
population_sd
```

```
## [1] 1.207779
```

The average `vote_average` in the full dataset is 6.05.

Step b

```
set.seed(123)
movies_sample <- movies2[sample(1:nrow(movies2), 25), ]
mean(movies_sample$vote_average)
```

```
## [1] 6.132
```

The average rating in the random sample of 25 movies is close to 6.05, but not the same. But this is normal because a small random sample will usually be different from the full population.

Step c

```
sample_means <- numeric(100)
for (i in 1:100) {
  s <- movies2[sample(1:nrow(movies2), 25), ]
  sample_means[i] <- mean(s$vote_average)
}
head(sample_means)
```

```
## [1] 5.884 6.088 5.892 5.856 6.300 5.528
```

We created 100 different random samples of 25 movies and calculated the average rating for each sample and most of these are still close to the population average of 6.05, but still not exactly the same.

Step d

```
summary(sample_means)
```

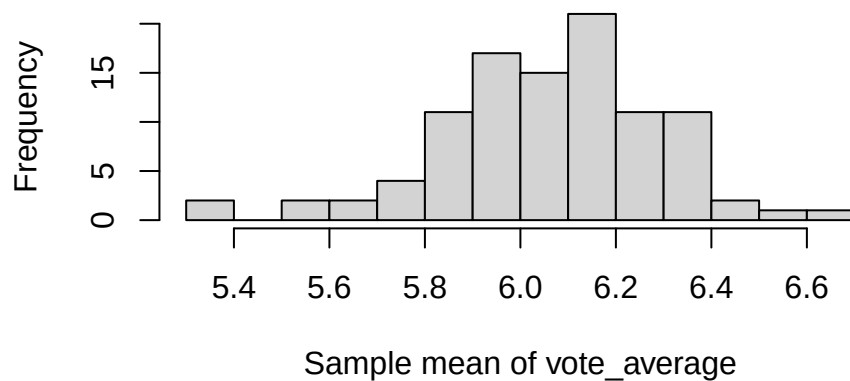
```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  5.380   5.916   6.090   6.065   6.209   6.632
```

```
sd(sample_means)
```

```
## [1] 0.2285405
```

```
hist(
  sample_means,
  main = "Distribution of Sample Means (n = 25, 100 samples)",
  xlab = "Sample mean of vote_average",
  ylab = "Frequency",
  breaks = 15
)
```

Distribution of Sample Means (n = 25, 100 samples)



The 100 sample means have an average that is close to the population average of 6.05. Most of the sample means are also close to 6.05, although they are not exactly the same. The standard deviation of the sample

means is about 0.23, showing that there is some variation between the samples. The histogram is roughly bell-shaped, which suggests that the sample means are approximately normally distributed.

Step e

The average of 25 movie samples should be nearly normally distributed by the Central Limit Theorem. Expected mean is 6.05 and the expected standard deviation is about 0.24. These values are closed to the one we got from the 100 random samples. Small differences are normal, because we only used 100 samples.

3. Is there a variable in your data set which you would expect to follow a discrete uniform distribution? Explain, without looking at the data, why you expect this data to be discrete uniform. If the data would indeed follow a discrete uniform distribution, what would you expect the mean and standard deviation of this variable to be? Next, plot a histogram of the variable, and plot an appropriate discrete uniform distribution on top of this histogram. Comment whether the variable indeed follows this expected distribution. If the distribution deviates from the discrete uniform distribution, explain why this is the case. **[6 points]**

A possible variable is `release_month`, because its values range from 1 to 12. If movies were released in any month with equal probability, then each month would have the same probability and this would be a discrete uniform distribution. The expected mean is 6.5 and the expected standard deviation is about 3.45.

```
k <- 12
exp_mean <- (1 + k) / 2
exp_sd <- sqrt((k^2 - 1) / 12)
exp_mean
```

```
## [1] 6.5
```

```
exp_sd
```

```
## [1] 3.452053
```

```
mean(movies2$release_month)
```

```
## [1] 6.735974
```

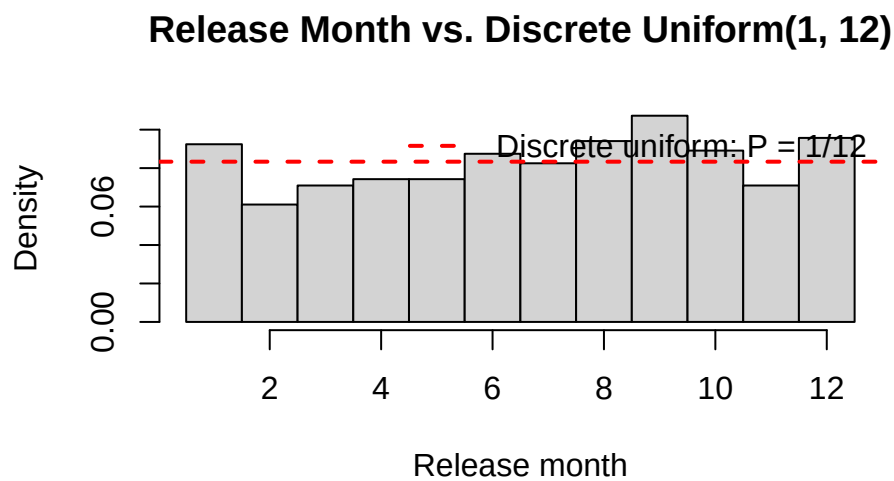
```
sd(movies2$release_month)
```

```
## [1] 3.441544
```

```
table(movies2$release_month)
```

```
##
##  1  2  3  4  5  6  7  8  9 10 11 12
## 56 37 43 45 45 53 50 57 65 54 43 58
```

```
hist(
  movies2$release_month,
  breaks = seq(0.5, 12.5, 1),
  freq = FALSE,
  main = "Release Month vs. Discrete Uniform(1, 12)",
  xlab = "Release month",
  ylab = "Density",
  col = "lightgray"
)
abline(h = 1/12, col = "red", lwd = 2, lty = 2)
legend("topright", legend = "Discrete uniform: P = 1/12",
      col = "red", lty = 2, lwd = 2, bty = "n")
```



In the actual data, the mean is 6.74 and the standard deviation is 3.44, which are close to the expected values. However, the histogram shows that the months are not exactly equally represented, so the variable does not perfectly follow a discrete uniform distribution. This could be because movies tend to be released in certain months for seasonal, holiday, and movie release reasons.

3 Week 3

For the next part of the assignment, assume that the movies in your data frame are a random sample of a larger population of movies.

1. Give a 90 percent confidence interval for the variable *revenue*. Interpret the result. [2 points]

```
ci_revenue <- t.test(movies2$revenue, conf.level = 0.90)$conf.int
ci_revenue
```

```
## [1] 73673117 96234088
## attr(,"conf.level")
## [1] 0.9
```

The 90% confidence interval for the population mean revenue is [\$73,673,117, \$96,234,088].

Since the population standard deviation is unknown, the sample standard deviation is used and the confidence interval is based on the Student's t-distribution. Because the sample is large ($n=606$), the Central Limit Theorem applies, meaning that the sampling distribution of the mean can be treated as approximately normal even though revenue itself is strongly right-skewed.

We are 90% confident that the true population mean revenue lies between \$73.7 million and \$96.2 million.

2

- a. How many movies in your sample were directed by a male director? How many by a female director? [2 points]
- b. Test the null hypothesis that the chance that a movie is being directed by a male or female director is equal. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Use an appropriate R function to test your hypothesis, and state your conclusion. Also give the p-value that corresponds to your test statistic. [2 points]
- c. Recode revenue such that it is measured in millions of dollars. What is the variance of revenue for movies with a male director? What is the variance of revenue for movies with a female director? [2 points]
- d. Assume that the variance of revenue for movies with a male director in your data is the same as the variance for movies with a male director in your population (that is, there is no sampling uncertainty in this estimate). Now, set up a hypothesis test to test for the null hypothesis that the variance of revenue for movies with a female director is equal to the variance for movies with a male director. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Calculate your test statistic and tell how your test statistic is distributed, and state your conclusion. Also give the p-value that corresponds to your test statistic. [2 points]

step a

```
gender_counts <- table(movies2$director_gender, useNA = "ifany")
gender_counts
```

```
##
## female    male    <NA>
##      50     522     34
```

In the sample, 522 movies were directed by men and 50 by women. Another 34 movies have no recorded value for `director_gender`, so they are excluded from the comparison. Among movies with a known director gender, about 91% were directed by men and 9% by women.

step b

```
n_male <- sum(movies2$director_gender == "male", na.rm = TRUE)
n_female <- sum(movies2$director_gender == "female", na.rm = TRUE)
n_total <- n_male + n_female

binom_test_res <- binom.test(x = n_male, n = n_total, p = 0.5)
binom_test_res

##
## Exact binomial test
##
## data:  n_male and n_total
## number of successes = 522, number of trials = 572, p-value < 2.2e-16
## alternative hypothesis: true probability of success is not equal to 0.5
## 95 percent confidence interval:
##  0.8863807 0.9344256
## sample estimates:
## probability of success
##           0.9125874
```

```
binom_test_res$p.value
```

```
## [1] 3.836994e-100
```

Among the 572 movies with a known director gender, 522 were directed by men and 50 by women. The sample proportion of male-directed movies is therefore about 91.3%.

The null hypothesis states that 50% of movies are directed by men, while the alternative hypothesis states that the proportion is different from 50%.

The binomial test gives a p-value of about 3.84×10^{-100} , which is much smaller than the 5% significance level. Therefore, the null hypothesis is rejected. There is strong statistical evidence that the proportion of movies directed by men is different from 50%.

step c

```
# recode revenue into millions of dollars
movies2$revenue_millions <- movies2$revenue / 1e6

# variance of revenue (in millions) by director gender
var_male <- var(movies2$revenue_millions[movies2$director_gender == "male"], na.rm =
  TRUE)
var_female <- var(movies2$revenue_millions[movies2$director_gender == "female"], na.rm =
  TRUE)

var_male
```

```
## [1] 28860.55
```

```
var_female
```

```
## [1] 4850.342
```

After converting revenue to millions of dollars, the variance is about 28,860.5 for male-directed movies and 4,850.3 for female-directed movies. This means the revenue of male-directed movies is much more spread out, with a variance about six times larger than that of female-directed movies.

step d

```
# treat the male variance as the known population variance (no sampling uncertainty)
sigma0_sq <- var(movies2$revenue_millions[movies2$director_gender == "male"], na.rm =
  TRUE)

female_revenue <- movies2$revenue_millions[movies2$director_gender == "female"]
female_revenue <- female_revenue[!is.na(female_revenue)]
n_female <- length(female_revenue)
s2_female <- var(female_revenue)

df_female <- n_female - 1
chi2_stat <- df_female * s2_female / sigma0_sq
chi2_stat
```

```
## [1] 8.235005
```

```
p_value <- 2 * min(pchisq(chi2_stat, df_female), 1 - pchisq(chi2_stat, df_female))
p_value
```

```
## [1] 1.437934e-11
```

The null hypothesis is that the revenue variance for female-directed movies is equal to 28,860.5, the variance for male-directed movies. The alternative hypothesis is that the two variances are different.

Using the 50 female-directed movies, the chi-square test statistic is approximately 8.24 with 49 degrees of freedom. The p-value is approximately 1.44×10^{-11} , which is much smaller than the 5% significance level.

Therefore, we reject the null hypothesis. There is strong evidence that the revenue variance for female-directed movies is different from that of male-directed movies. Since the female-directed movies have a much smaller variance, their revenues are less spread out than those of male-directed movies.

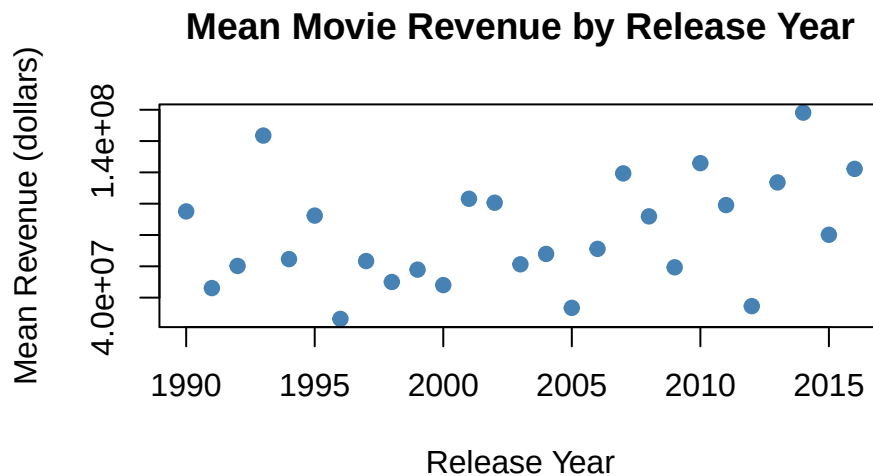
3

- Create a scatter plot with year on the x axis and mean of revenue within that year on the y-axis. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Is revenue of movies increasing or decreasing over time? **[3 points]**
- Recreate the scatter plot with year on the x axis and mean_revenue on the y-axis, but now add bars around each point, indicating the 95% confidence interval. **[3 points]**
- If you look closely at the resulting plot in b.), you will probably see that the confidence intervals in some years are wider than in other years. What are some possible reasons that a confidence interval in a given year can be wide? Can you verify this using your data? **[4 points]**

step a

```
yearly_revenue <- aggregate(revenue ~ release_year, data = movies2, FUN = mean)

plot(
  yearly_revenue$release_year, yearly_revenue$revenue,
  main = "Mean Movie Revenue by Release Year",
  xlab = "Release Year",
  ylab = "Mean Revenue (dollars)",
  pch = 19,
  col = "steelblue"
)
```



The scatter plot shows that average movie revenue changes quite a lot from year to year, but there is a slight upward trend over time. Revenues are generally lower in the earlier years and somewhat higher in the later years.

The points are widely spread around this trend, so the relationship between release year and average revenue is fairly weak. Overall, average revenue seems to increase over time, but not in a smooth or consistent way.

step b

```
# per-year mean, standard error, and 95% CI half-width
year_stats <- aggregate(
  revenue ~ release_year, data = movies2,
  FUN = function(x) {
    n <- length(x)
    m <- mean(x)
    se <- sd(x) / sqrt(n)
    ci_halfwidth <- qt(0.975, df = n - 1) * se
    c(mean = m, lower = m - ci_halfwidth, upper = m + ci_halfwidth)
  }
)
year_stats <- do.call(data.frame, year_stats)
names(year_stats) <- c("release_year", "mean_revenue", "lower", "upper")

plot(
  year_stats$release_year, year_stats$mean_revenue,
```

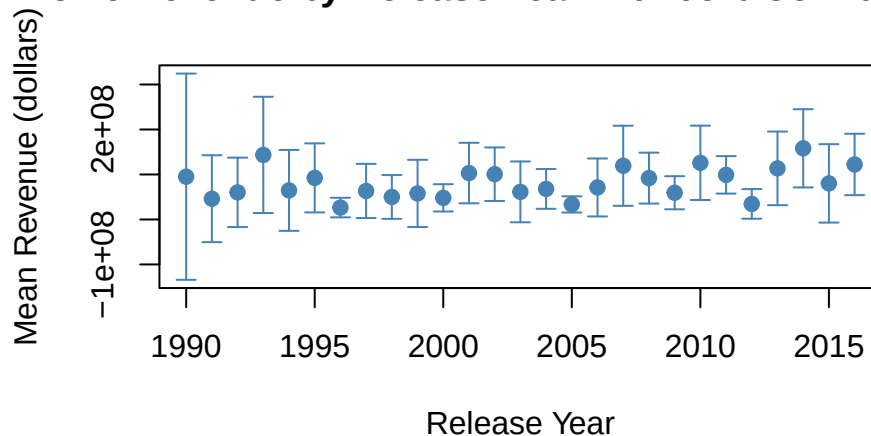
```

main = "Mean Movie Revenue by Release Year with 95% Confidence Intervals",
xlab = "Release Year",
ylab = "Mean Revenue (dollars)",
pch = 19,
col = "steelblue",
ylim = range(c(year_stats$lower, year_stats$upper))
)

arrows(
  x0 = year_stats$release_year, y0 = year_stats$lower,
  x1 = year_stats$release_year, y1 = year_stats$upper,
  angle = 90, code = 3, length = 0.05, col = "steelblue"
)

```

Movie Revenue by Release Year with 95% Confidence



Adding 95% confidence intervals shows that the yearly mean revenues are estimated with quite different levels of precision. In years with only a few movies, such as 1990 and 1991, the intervals are very wide. In later years, where more movies are included, the intervals become much narrower and the estimates are more precise.

The confidence intervals also make the upward trend look less clear than in the original scatter plot. Many of the intervals overlap, so the differences between individual years are often not statistically clear. Overall, later years still tend to have higher mean revenues, but there is a lot of uncertainty and year-to-year variation.

step c

```

year_stats <- aggregate(
  revenue ~ release_year, data = movies2,
  FUN = function(x) {
    n <- length(x)
    sdev <- sd(x)
    se <- sdev / sqrt(n)
    tcrit <- qt(0.975, df = n - 1)
    c(n = n, sd = sdev, se = se, width = 2 * tcrit * se)
  }
)

```

```

year_stats <- do.call(data.frame, year_stats)
names(year_stats) <- c("release_year", "n", "sd", "se", "width")
year_stats[year_stats$release_year %in% c(1990, 2005, 2015), ]

```

```

##   release_year  n      sd      se    width
## 1         1990  5 184579521 82546471 458371491
## 16         2005 28  46374546  8763965  35964344
## 26         2015 35 253547492 42857348 174193222

```

```

# correlate CI width with each candidate driver
cor(year_stats$width, year_stats$n)

```

```
## [1] -0.4088846
```

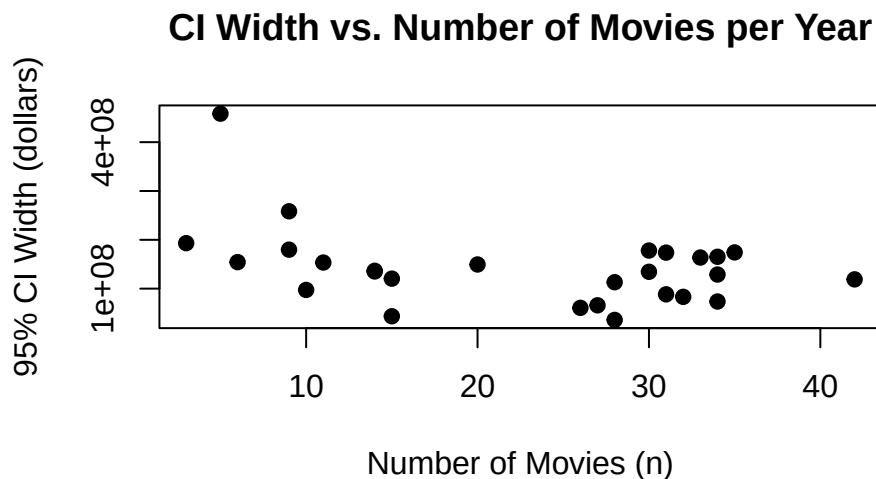
```
cor(year_stats$width, year_stats$sd)
```

```
## [1] 0.4448869
```

```

plot(year_stats$n, year_stats$width,
     main = "CI Width vs. Number of Movies per Year",
     xlab = "Number of Movies (n)", ylab = "95% CI Width (dollars)", pch = 19)

```



The width of a yearly confidence interval depends mainly on two things: the number of movies in that year and how much their revenues vary.

A smaller sample size gives a wider confidence interval because the standard error becomes larger. The t critical value is also larger when the sample size is small, which makes the interval even wider. A larger standard deviation also increases the width because revenues within that year are more spread out.

This can be seen in the data. For example, 1990 has a very wide confidence interval because it only contains 5 movies and revenue varies a lot between them. In contrast, 2005 has a much narrower interval because it has more observations and a relatively low standard deviation.

Sample size is therefore not the only factor. For example, 2015 has 35 movies, but its confidence interval is still fairly wide because revenue varies greatly within that year. This shows that both sample size and the variability of revenue affect the width of a confidence interval.

4 Week 4

1. There is an argument going on in the movie studio. *Bob* claims that production budgets are getting out of hand, and that the studio should focus on making cheaper movies. *Chantal* disagrees. She tells Bob that “Every dollar we spend on movie production is more than offset by the increase in movie revenue”.

- a. Set up a regression model to test Chantal’s claim, and estimate it. That is, estimate:

$$\text{Revenue}_i = \beta_0 + \beta_1 \text{Budget}_i + \varepsilon_i.$$

Print a summary of your estimated model. **[2 points]**

- b. What is the estimated value of β_1 and how do you interpret it? **[2 points]**
- c. Test for the null hypothesis that $\beta_1 \leq 1$. Report the p-value and state your conclusion. **[2 points]**
- d. Next, estimate the model

$$\text{Log Revenue}_i = \beta_0 + \beta_1 \text{Log Budget}_i + \varepsilon_i.$$

When creating the variables Log Revenue and Log Budget, make sure that movies with a Revenue or Budget of zero are assigned the value “NA”. Print a summary of your estimated model. **[2 points]**

- e. What is the estimated value of β_1 and how do you interpret it? **[2 points]**
- f. Which model has better fit? The level-level model or the log-log model? Explain. **[2 points]**
- g. Who do you think is correct? Bob or Chantal? What would you advise the movie studio to do? **[2 points]**

step a

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

step b

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

step c

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

step d

```
#WRITE YOUR CODE HERE
```

Your Answer:

Write your formulated response here.

step e

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step f

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step g

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

2

- a. Make a scatterplot with log revenue on the y-axis, and log budget on the x-axis. Within the same scatterplot, draw a straight line which summarizes the association between the log of revenue and log of budget. [3 points]
- b. Using the insights from the scatterplot, your linear model, and your data, find an example of a movie that vastly overperformed. That means, find a movie that had much more revenue than expected, given its budget. For this movie, report its log of revenue, its predicted log of revenue, and its residual. [3 points]

step a

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step b

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

5 Week 5

- a. Make a frequency table of the variable genre. Which are the three genres that occur the most? [2 points]

- b. Create a new variable called genre2. Make sure that the three most popular genres remain the same, but that all the other genres get categorized into one genre called “Other”. Report a frequency table of this new variable genre2. **[2 points]**
- c. Estimate an OLS model which has as dependent variable the log of revenue of a movie, and as independent variable the log of budget, a dummy for each level that genre2 can take, and the year of release. Show a summary of the resulting model and interpret each coefficient. **[4 points]**

The movie studio that you work at is releasing a new movie in 2026. It will be a Drama movie with a budget of 20,000,000. It will have a runtime of 120 minutes.

- d. Estimate a model that is able to predict the revenue of this movie. Give its predicted revenue and include a 90% prediction interval. **[5 points]**
- e. Plot the residuals against various independent variables in the model. Do you find any evidence of heteroskedasticity? **[2 points]**
- f. One executive at the studio wants to time the release of the movie to a specific month of the year such that they can maximize revenue. Another executive is of the opinion that the month in which the movie is released will not have a big impact on its revenue. Estimate a model that can test whether the month of release impacts the revenue of a movie. Use an appropriate hypothesis test to test whether month of release matters. What would you advise the movie studio regarding the timing of the release of the movie? **[5 points]**

step a

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step b

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step c

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step d

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step e

#WRITE YOUR CODE HERE

Your Answer:

Write your formulated response here.

step f

#WRITE YOUR CODE HERE